S Vasist Muktinutulapati

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.90
FIITJEE Junior College	Class XII (Telangana State Board of Intermediate Education)	2020	94.70%
Sarathi School	Class X (Central Board of Secondary Education)	2018	89.40%

WORK EXPERIENCE AND INTERNSHIPS

Pacific Surfaces - Hosur

R&D and Post Production

- **Quality Assurance:** Managed **post-production quality control** and **surface defect inspection** for **engineered quartz slabs** to ensure mechanical specifications.
- **R&D Support:** Assisted in evaluating **resin-aggregate binding**, **pigment consistency**, and slab curing parameters to reduce finishing defects.
- **Process Oversight:** Monitored post-curing finishing lines to identify **bottlenecks** and maintain **product quality** standards.

INTERNSHIP , National Chemical Laboratory (NCL)

Risk Assessment for Chemical Process Industries (RACPI) .

- **Objective** : Preventing and Controlling **runaway reactions** .

- Collection of **reaction database**

- Thermodynamic aspect : Calculated **MTSR** ,**Tad** for different reactions to find the runaway criteria .

- Kinetic Aspect : Worked on the application of **RTDs** in preventing and controlling the runaway reactions .

- Criteria for runaway limits : Worked on **divergence criteria** , implemented on **MATLAB** , to check the feasibility of the reaction

SUMMER INTERNSHIP , MANIKGARH CEMENT

Alternate Fuels for Cement Industry

- **Objective** : Use of **alternate fuels** in cement industry

- Types: Worked on different types of alternate fuels .

- **Hydrogen Fuel**: Application of Hydrogen Fuel in Cement Industries .

- Worked on the advantages and disadvantages of **Hydrogen Fuel** as a **Alternate Fuel**

PROJECTS

Master's Thesis Project

Photocatalytic Synthesis of Hydroquinone using Milli-Fluidic Reactors

- Synthesizing and Characterizing **ZnO** and **Au doped ZnO** as catalyst.
- Performing the reaction in **microreactor** and **millifluidic reactor** using **photocatalysis** mechanism.

- Evaluating and comparison of **selectivity** and **yield** of **hydroquinone** formed.

B.E. Project

Fabrication of EPDM-Nanocellulose composite for Oil-Water Separation

- Prepared **EPDM** and **Nanocellulose** solution with different compositions on glass slides and metal mesh .
- Dispersed **nanocellulose** in **toluene** solution using **bath-sonicator** .
- Coated the solution when nanocellulose gets dispersed in the solution .
- Analyzed results using **Optical Microscope** and **Goniometer** .

Machine Learning in Process System Engineering

- Developed **Artificial Neural Network** (**ANN**) model to predict **Overall Heat Transfer Coefficient**
- Applied **K-Means clustering** on **flow velocity** and **heat transfer coefficient** data, utilizing the **elbow method** to determine the optimal number of clusters.
- Validated ML regression models against experimental transport data, achieving **high predictive accuracy** with an **R2 score** of **0.96**.

Multi Objective Optimization in Cement Industry

- Developed an **ANN** model for **compressive strength** and **energy consumption**.
- Implemented **NSGA-II** algorithm to minimize energy consumption and maximize compressive strength.
- Evaluated **Pareto front** for trade off analysis finding optimum range.

CERTIFICATIONS

- Completed Foundational Level in **BS Degree** in **Programming and Data Science** from **IIT- Madras**

SKILLS

Technical Skills: Aspen Plus, Aspen HYSYS, ANSYS, MATLAB, Simulink, DWSIM, C Programming, Python, Julia

Soft Skills: Leadership, Management, Teamwork, Problem-Solving, Logical and Creative Thinking, Communication

RELEVANT COURSES

Heat Transfer, Mass Transfer, Chemical Reaction Engineering, Fluid Mechanics, Thermodynamics, Advanced Process Control, Chemical Technology, Mechanical Operations, Plant Design, Reactor Modelling, Machine Learning for Process System Engineering, Optimization Techniques, Microfluidics, Photocatalysis, Advanced Numerical Methods, Process Integration, Sustainability in Chemical Engineering

POSITIONS OF RESPONSIBILITY

Teaching Assistant

- **Heat Transfer(Theory)** : Evaluated examination answer scripts and coursework assignments for the undergraduate **Heat Transfer** course.
- **Heat Transfer(Lab)** : Explained the core principles and guided students in operating **heat exchanger** setups and collecting the data.
- **Fluid Mechanics (Lab)** : Instructed students on fundamental fluid mechanics experiments, focusing on **hydrostatic properties** and **fluidized bed characteristics**.

EXTRACURRICULAR

Cricket: Played for **BITS Cricket Team** and **U-19 team.**

Tennis: Played **TSTA** tournament.

Playing **Tabla** and **Flute.**

Playing **Badminton, Squash, Table Tennis, Carrom Board, Chess.**